

# Smart home

Revise 360 lesson plan — 60 minutes

|                 |                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Unit            | OCR J277 1.1 Systems architecture — Paper 1, Computer systems                                                     |
| Lesson          | Smart home (lesson 4 of the unit)                                                                                 |
| Lesson ID       | sa-104                                                                                                            |
| Experience ID   | sa-104                                                                                                            |
| Plan ID         | PLAN-SA-L04                                                                                                       |
| Content version | 6a7f647eda98                                                                                                      |
| Website         | <a href="https://revise360.co.uk/experience.html?id=sa-104">https://revise360.co.uk/experience.html?id=sa-104</a> |

## Big question, outcomes and success criteria

**Big question** — What are embedded systems, and what are their characteristics?

### Prerequisites

The lesson before this one is sa-103. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: sa-101, sa-102.

### Assessable outcomes

1. **sa-104-o1** — understand what is meant by the term embedded system
2. **sa-104-o2** — know several examples of embedded systems
3. **sa-104-o3** — understand how an embedded system takes input, processes it and produces output

## Vocabulary, resources and preparation

**Key vocabulary** — Von Neumann architecture, MAR

### Resources

|                   |                                            |
|-------------------|--------------------------------------------|
| SLIDES            | presentations/SA_L04_SmartHome_Lesson.pptx |
| WORKSHEET         | worksheets/SA_L04_SmartHome_Worksheet.docx |
| ONLINE EXPERIENCE | experience.html?id=sa-104                  |

### Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

### How pupils get in

1. Go to `topics.html` on revise360.co.uk.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.

3. Choose the topic, then this lesson's card.

**Worth knowing before you promise anything.** In this build APP\_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

## Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

| MIN   | PHASE                                     | TEACHER                                                                                      | PUPILS                                                                          | EVIDENCE TO CHECK                                                                      |
|-------|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0-5   | <b>Retrieve</b>                           | Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites. | Answer the starter on the worksheet.                                            | The starter box on SA_LO4_SmartHome_Worksheet.docx is filled in.                       |
| 5-8   | <b>Review the starter</b>                 | Take answers; correct the one most got wrong.                                                | Correct their own starter.                                                      | A correction written, not just a tick.                                                 |
| 8-18  | <b>Explain and model</b>                  | The new idea in two or three steps, with the example worked in front of the class.           | Listen and answer questions about the example.                                  | They can say why, not only what.                                                       |
| 18-21 | <b>Set up and settle</b>                  | Sign-in, launch, and what to write before anyone taps an answer.                             | Sign in, open the lesson, find What is an embedded system?.                     | Everyone is in and on station 1.                                                       |
| 21-43 | <b>Online: work the stations in order</b> | Pupils read each wall, write the key fact and the challenge, then answer at the badge.       | Read each wall, write the key fact and the challenge, then answer at the badge. | The worksheet has writing on it for each of the 6 stations before the badge is tapped. |
| 43-48 | <b>Check understanding</b>                | The hinge question, hands down.                                                              | Answer the hinge question, hands down.                                          | The split across the options.                                                          |
| 48-56 | <b>Apply</b>                              | Exam practice on the worksheet, in silence.                                                  | Exam practice, in silence.                                                      | worksheet, Exam practice a) to c)                                                      |
| 56-60 | <b>Exit and set up the next lesson</b>    | The exit question and the retrieval link to next lesson.                                     | Write the exit answer.                                                          | One sentence that answers the big question.                                            |

## The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

### 1. What is an embedded system?

- A computer system built into a larger device.
- It has a dedicated function: it does one job, or a small set of jobs.
- Examples: washing machines, microwaves, traffic lights and car engine management.

**Challenge on this wall:** Explain what an embedded system is.

### 2. Characteristics

- Limited functions: runs one dedicated program.
- Usually small, low power and cheap to mass produce.
- Very reliable, and often has to respond in real time.

**Challenge on this wall:** Why are embedded systems usually cheap and low power?

### 3. Embedded systems everywhere

- At home: washing machines, microwaves, smart thermostats, TVs and fridges.
- Out and about: traffic lights, cash machines and car systems, like automatic lights and anti-lock brakes.
- On you: fitness trackers and smartwatches.

**Challenge on this wall:** List five embedded systems you have used today.

### 4. Input, process, output

- Input: buttons, dials and sensors, such as a water temperature sensor.
- Process: a microcontroller runs the program and makes decisions.
- Output: motors, heaters, lights and displays.

**Challenge on this wall:** Choose a device in your home and describe its inputs, its processing and its outputs.

### 5. Programming embedded systems

- The program is stored permanently in the device, often called firmware.
- Embedded systems are often programmed in C or C++, which run fast and give close control of the hardware.
- Some firmware can be updated, like a smart TV's software update.

**Challenge on this wall:** What programming language is mainly used for embedded systems, and why?

### 6. Embedded or general purpose?

- A general-purpose computer, like a laptop, can run many different programs.
- An embedded system is built in to one device and does one dedicated job.
- Some devices blur the line, like smart TVs that run apps.

**Challenge on this wall:** Explain why a laptop is not an embedded system.

## Retrieval and the check for understanding

**Starter** — List five devices in your home that contain a computer but don't look like one. What does each one do with the data it collects?

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

### Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (sa-104-s3-c2), so the wording a pupil sees matches the wording you say.

**Complete the exam sentence: embedded systems have limited...**

**Correct:** functions

### What a pupil might choose instead

- colours
- keyboards
- screens

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

**The course's own response to a wrong answer here:** Embedded systems have limited functions and are built into a larger machine.

**One response, not four.** one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

## Guided to independent practice

|           |    |                                                                                                                           |
|-----------|----|---------------------------------------------------------------------------------------------------------------------------|
| Recognise | 10 | The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.                   |
| Construct | 2  | Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer. |

### Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

### Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

## Delivering the online experience

### Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

### The stations, and what to record at each

| #     | STATION                                   | ACTIVITIES | RECORD ON PAPER                                                     |
|-------|-------------------------------------------|------------|---------------------------------------------------------------------|
| 1     | What is an embedded system?               | 2          | An embedded system is...                                            |
| 2     | Characteristics                           | 2          | Embedded systems are usually...                                     |
| 3     | Embedded systems everywhere               | 2          | Examples of embedded systems include...                             |
| 4     | Input, process, output                    | 1          | A washing machine is a computer because...                          |
| 5     | Programming embedded systems              | 2          | Embedded systems are programmed using...                            |
| 6     | Embedded or general purpose?              | 2          | The difference between embedded and general-purpose computers is... |
| final | Final challenge: spot the embedded system | 1          | the answer to the station's challenge                               |

### Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

## Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

## Assessment, evidence and unfinished work

| ID        | QUESTION                                                               | MARKS | WHERE                                                  |
|-----------|------------------------------------------------------------------------|-------|--------------------------------------------------------|
| sa-104-x1 | Name an embedded system and explain why it is described as a computer. | 3     | worksheet, Exam practice a); deck, Exam practice slide |
| sa-104-x2 | Describe two characteristics of embedded systems.                      | 4     | worksheet, Exam practice b); deck, Exam practice slide |
| sa-104-x3 | Explain why a laptop is not an embedded system.                        | 2     | worksheet, Exam practice c); deck, Exam practice slide |

## Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/sa-104.json`, which is not published with the website.

## Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is an embedded system?** — A laptop used for many different tasks; A program saved in the cloud
- **Which is an example of an embedded system?** — A desktop PC; A laptop
- **Why are embedded systems usually cheap to make?** — They have no processor; They are made by hand
- **Which of these contains an embedded system?** — A pencil; A paper book
- **Complete the exam sentence: embedded systems have limited...** — colours; keyboards

## Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

## Exit, next lesson and homework

### Exit check

One of these, written, not a show of hands:

- What an embedded system is
- Examples of embedded systems
- How embedded systems take input, process and output

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

### Retrieval link to next lesson

Next is `sa-105-test`, End-of-topic test. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

### Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

## Shorter lessons, longer lessons and no internet

### A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

### **If the internet or the devices are unavailable**

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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